

**Institute of Technology of Cambodia
International Program Year 3 Generation 1**



**Topic 4: Hybrid Cloud & Remote Management
Class: Network System Design**

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1. Project Objective

1.1. Network Overview

This project simulates a real-world hybrid cloud scenario for a tech startup with a main Office site and a remote Data Center. The two sites are connected over the public internet, simulated using an "Internet" router in Cisco Packet Tracer to represent WAN separation.

To ensure secure communication of sensitive data between the Office LAN (192.168.10.0/24) and Data Center LAN (10.10.10.0/24), we configured a Site-to-Site IPsec VPN tunnel between the Office Router and Data Center Router. This encrypts only corporate traffic (split tunneling), while general internet access remains unencrypted for efficiency.

For remote management and operations (Ops), we enabled SNMP (v2c) on the Office Router and deployed a Syslog & SNMP server in the Data Center. This allows centralized monitoring and instant alerts if a device or link fails (e.g., interface shutdown).

The design demonstrates key WAN concepts: secure connectivity over public internet, bandwidth optimization, and proactive network monitoring to minimize downtime.

1.2. The Problem

Modern enterprises require secure communication between a Branch office and a Data Center. Transmitting sensitive data over the public internet without encryption exposes the company to "Man-in-the-Middle" attacks and data breaches. Additionally, administrators at the Data Center need a real-time alert into the health of remote hardware to minimize downtime.

1.3. The Solution

We implemented a **Site-to-Site IPsec VPN** using Cisco ISR 2911 routers. This solution provides:

- **Confidentiality:** AES Encryption ensures data are encrypted and cannot be read by outsiders.
- **Integrity:** SHA Hashing ensures data is not tampered with in transit.
- **Centralized Monitoring:** SNMP and Syslog protocols allow the Data Center Server to track router status and receive instant alerts.
- **Split Tunneling:** Optimizes bandwidth by only encrypting traffic destined for the corporate network.

2. IP Addressing Scheme Security Implementation

Device	Interface	IP Address	Subnet Mask	Default Gate way
Office PC	NIC	192.168.10.10	255.255.255.0	192.168.10.1
Office Router	Gig0/0 (LAN)	192.168.10.1	255.255.255.0	N/A
Officer Router	Gig0/1 (WAN)	203.0.113.1	255.255.255.252	203.0.113.2
Internet Router	Gig0/0	203.0.113.2	255.255.255.252	N/A
Internet Router	Gig0/1	198.51.100.1	255.255.255.252	N/A
DC Router	Gig0/0 (WAN)	198.51.100.2	255.255.255.252	198.51.100.1
DC Router	Gig0/1 (LAN)	10.10.10.1	255.255.255.0	N/A
DC Server	NIC	10.10.10.10	255.255.255.0	10.10.10.1

No VLANs were required for this basic setup.

3. VPN Design (Why IPsec)

We chose IPsec for the site-to-site VPN because it is a robust, industry-standard protocol suite specifically designed for securing IP communications over untrusted networks like the public internet.

Key reasons for selecting IPsec:

- **Confidentiality:** Uses strong encryption (AES) to protect data from eavesdropping or man-in-the-middle attacks.
- **Integrity:** Employs hashing (SHA) to detect any tampering during transit.

- **Authentication:** Supports pre-shared keys (or certificates in production) to verify endpoints.
- **Site-to-Site Suitability:** IPsec operates at the network layer (IP), making it transparent to end devices and ideal for connecting entire networks (Office ↔ Data Center) rather than individual users.
- **Standards-Based and Vendor-Neutral:** Widely supported on Cisco routers (e.g., ISR 2911), scalable, and proven for enterprise WAN connectivity.

Compared to alternatives like SSL VPN (better for remote access), IPsec excels in site-to-site scenarios with full network access, high performance, and strong security for bulk data transfer.

We used IKE Phase 1 (ISAKMP policy with AES, SHA, pre-share, group 2) for secure key exchange and Phase 2 (transform-set with ESP-AES + ESP-SHA-HMAC) for data encryption, applied via crypto map on WAN interfaces.

4. Verification & Testing

4.1. VPN Tunnel Verification

Split tunneling is configured so that only traffic between the Office LAN and Data Center LAN is encrypted and routed through the IPsec VPN tunnel. All other traffic (e.g., to the public internet) exits directly via the Office Router's WAN interface without encryption.

This is achieved by defining "interesting traffic" in an ACL (access-list 100 permit ip 192.168.10.0 0.0.0.255 10.10.10.0 0.0.0.255) matched in the crypto map — only matching packets trigger the tunnel.

The command `show crypto isakmp` was used to verify the tunnel status. The output showed QM_IDLE, confirming the tunnel is active and encrypted.

```
Office-Router#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id slot status
198.51.100.2 203.0.113.1  QM_IDLE        1061      0  ACTIVE

IPv6 Crypto ISAKMP SA
```

4.2. Split Tunneling Test

- **Result:** A ping from Office PC to 10.10.10.10 increased the #pkts encaps counter on the router.
- **Result:** A ping to the public 203.0.113.2 did not increase the counter, proving that non-corporate traffic bypasses the VPN.

4.3. Syslog Failure Test

When the Office Router's LAN interface was manually shut down, the Data Center Server immediately logged the event:

%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down

This confirms that the management system is functioning correctly over the secure tunnel.

5. Monitoring (SNMP + Syslog)

To enable real-time remote management from the Data Center:

- **SNMP (v2c):** Configured on the Office Router with snmp-server community public RO and snmp-server host 10.10.10.10 public. This allows the DC Server (running SNMP service) to poll device status, interface health, and performance metrics.
- **Syslog:** Configured with logging host 10.10.10.10, logging source-interface g0/0, and logging trap warnings (or similar level). Syslog messages (e.g., interface changes, errors) are sent to the DC Server for centralized logging.

The DC Server runs both Syslog and SNMP services, providing a single point for monitoring and alerting. This setup ensures the IT team receives immediate notifications of issues over the secure VPN tunnel.

Note to instructor:

- All router and switch passwords have been set to cisco.
- The enabled secret password is set to class.

- To trigger the VPN, please initiate a ping from the **Office PC** to the **Data Center Server (10.10.10.10)**.

```

Router0
Office-Router#show crypto ipsec sa

interface: GigabitEthernet0/1
  Crypto map tag: MYMAP, local addr 203.0.113.1

protected vrf: (none)
local  ident (addr/mask/prot/port): (192.168.10.0/255.255.255.0/0/0)
remote ident (addr/mask/prot/port): (10.10.10.0/255.255.255.0/0/0)
current_peer 198.51.100.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 14, #pkts encrypt: 14, #pkts digest: 0
  #pkts decaps: 13, #pkts decrypt: 13, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 1, #recv errors 0

  local crypto endpt.: 203.0.113.1, remote crypto endpt.:198.51.100.2
  path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
  current outbound spi: 0xE3468AA2(3813051042)

inbound esp sas:
  spi: 0x31623E91(828522129)
--More--

  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 18, #pkts encrypt: 18, #pkts digest: 0
  #pkts decaps: 17, #pkts decrypt: 17, #pkts verify: 0
  #pkts compressed: 0, #pkts decompressed: 0

```

6. Failure Test Evidence

To prove monitoring functionality:

- Manually shut down the Office Router's LAN interface (Gig0/0): interface GigabitEthernet0/0 → shutdown.
- **Result:** The DC Syslog Server immediately received the alert log:
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down

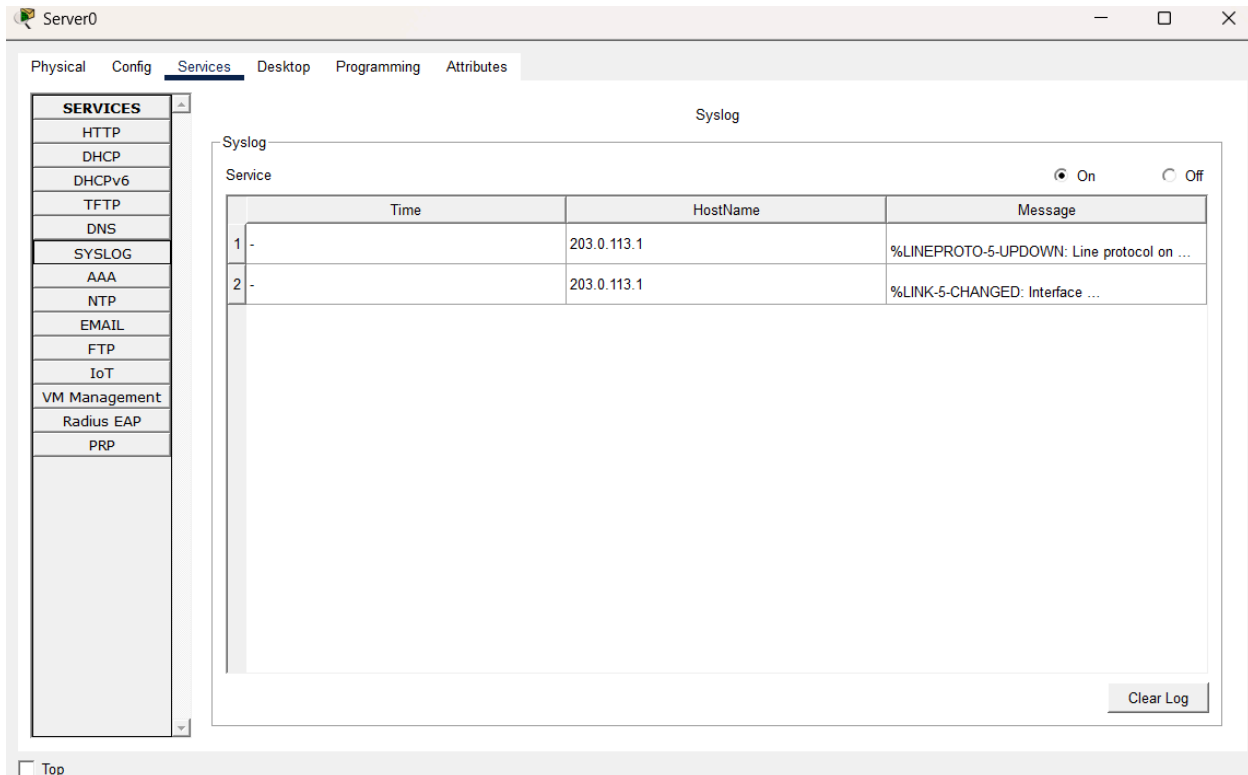


Figure: Syslog server receiving real-time interface status alerts from Branch Office via the IPsec Tunnel (when trigger the failure)

Syslog			On	Off
Service				
Time	HostName	Message		
1 -	203.0.113.1	%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to...		
2 -	203.0.113.1	%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up		
3 -	203.0.113.1	%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to...		
4 -	203.0.113.1	%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to administratively down		

Figure: Syslog server receiving real-time interface status alerts from Branch Office via the IPsec Tunnel (when network is restored)

This confirms syslog messages traverse the VPN tunnel successfully, and the management server detects and logs failures in real time. SNMP would also reflect the interface down status if polled.

7. Conclusion

This project successfully implements a secure, efficient hybrid cloud connection using site-to-site IPsec VPN with split tunneling, combined with centralized SNMP and Syslog monitoring. It addresses the core requirements: encrypted

site-to-site communication over public internet, bandwidth optimization, and immediate alerting for device failures.

The design is practical, scalable, and aligns with real-world enterprise WAN and Ops practices. We chose IPsec for its strong security features and split tunneling for performance gains. The simulation (.pkt file) is fully functional (passwords: cisco / class), and all tests (VPN tunnel, split tunneling, failure logging) pass successfully.

This solution enhances data protection, reduces downtime through proactive monitoring, and demonstrates key networking skills in WAN security and remote management.